

Evolution of the Competencies to Embrace Digital Technology for Sustainable Development

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ABSTRACT

The paper aims to highlight the way how competencies are changing, following technological progress and new tasks for sustainable development. New social challenges demand a new generation of competencies to apply technologies to produce positive effects on the wellbeing of society. There are known approaches to describing the competencies based on the assumption that they are applicable to any industry, but they are not. We suggest considering the quantity and quality of competencies in the context of industry, identifying the special abilities for its improvement with help of digital technologies. The industrial transformation promotes sustainability if it is based on a deep understanding of technology and industry. The result of this research is the designed concept of competencies for digital transformation based on a variety of industries. This is a preliminary step for advancing the existing concepts of digital competencies to transform the industry to support sustainable development.

CCS CONCEPTS

• **Competence**; • **Digital Technology Use**; • **Social development**;

KEYWORDS

Digital transformation, Knowledge and Skills, Digitalization of Society, Digital Literacy

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1 INTRODUCTION

The progress in digital technologies is promising for the improvement of wellbeing of persons, business entities and society as a whole [78]. Many initiatives from private companies up to national strategies are promoting the concept of digital transformation of

industries and business [44, 50]. The main reason for this is the fact that modern digital technologies are contributing to the sustainable development of society [11]. The digital technologies, as next-generation IT, are becoming the major engine of growth in present-day society, and access to them is a prerequisite for its economic development.

The success of digital transformation depends on a system of factors that generates the growth potential [25, 54], while the competencies bring all these factors together into each case. The purpose of this study is to design the concept of the competencies to transform industries with the help of digital technologies as a basis for sustainable development. The society needs competence that ensures effective implementation and usage of digital technologies. For the sake of brevity, let us call these digital competencies (DC). The object of research is the conceptualization of the competencies necessary for sustainability in the digital age.

In the context of this study, by digital technologies, we mean modern general-purpose or emerging technologies, such as big data, artificial intelligence, neural network technology, blockchain technologies, fog computing, quantum computing, advanced manufacturing, industrial Internet, sensor technologies and components in robotics, wireless communications, augmented and virtual reality, etc. In some publications, the concept of digital technologies (DT) includes personal computers, software, and other technologies engineered in the 20th century [75]. However, the development and improvement of the DT have evolved to the fundamentally new scientific and technical level in terms of access to big data, computational performance, or new and improved electronic participating databases. DT are also known as general-purpose technology [13, 32].

The term “sustainable development” is interchangeable with other concepts like “green economy” or “sustainable growth” and covers the issues from philosophy to materials science [9]. The recognized idea that sustainable development “meets the needs of the present without compromising the ability of future generations to meet their own needs” [15]. UNESCO considers sustainability as a paradigm for thinking about the future in which environmental, societal, and economic considerations are balanced in the pursuit of an improved quality of life [72, 73]. There are a lot of ways for business entities and countries to reach sustainable development which are reflected in the various aspects of the sustainable development goals approved by many national governments and international organizations [62].

The widespread technologies among society become user-friendly and users constantly gain experience. As a result, they require less training to possess the competencies to apply technologies to everyday or industry tasks. Society needs competencies that

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ensure the effective implementation of DT in social and business activities to significantly improve them with the help of the so-called digital transformation [1, 65]. It is essential to create institutional arrangements for the buildout and development of a new generation of digital competencies. The diversity of DT and the rate of change of society's technological landscape indicate that the competencies and requirements for them are characterized by specific properties, particularly a high update rate. The challenge to determine the essence, structure, and characteristics of digital competencies is the top priority for the formulation of methodological provisions for increasing the efficiency of interaction between human and DT interactions.

1.1 Methods and Materials

The theoretical part of research is based on materials of scientific publications and reports. Sources of factual data for empirical methods include the analytical reports of international organizations [58, 78], industries reports [79], and row data from World Intellectual Property Institute [77]. The essence of competencies and requirements is being transformed following the development of DT and lines of business involved in sustainable development.

The research on the competencies needed by practitioners in the context of digitalization should be cross-disciplinary, including a theoretical and methodological framework of the following scientific fields:

- applied information science in economics and management as related to business and economic transformation through DT [28, 40],
- the educational system as a traditional institution for developing and promoting competencies [12, 64, 66],
- knowledge management in terms of exchange and transfer of knowledge and competencies [2, 30, 63],
- innovation management in terms of innovation activities and creating innovative solutions as a mechanism for gaining competitive advantage from the process of the digital transformation of society [18, 26, 74].

The methods used in this study include systems analysis, generalization, and semantic modeling. The study's informational sources and materials comprise data from official sources, state administrative bodies, and international organizations, including research reports of companies and expert communities.

1.2 Review of approaches for understanding of digital competencies

Recent research incorporates several approaches to identifying the quantity and quality of competencies required for sustainable development of society in a digital age. The approaches considered below address the competencies based on different views: standardization of competence, labor market, professional activity, digital technologies, and the forecast for the future.

The utilitarian view on digital competencies defines the requirements for competencies needed for work with specific DT, contributing to the emergence of computer, information, and digital literacy [45, 46, 49]. This approach more frequently receives support from international organizations [14, 27, 47, 61, 75], since its application

allows educators to establish a system of multifunctional competencies that is beneficial to each practitioner and contributes to the economic success of society. The European Commission promotes the digital competence framework "DigComp 2.0", consisting of five dimensions: information and data literacy, communication and collaboration, digital content creation, safety, and problem-solving [19]. The OECD [58] identifies the transformative competencies that help shape a better future, including creating new value, reconciling tensions and dilemmas, and taking responsibility. General terms in descriptions of competencies provide many opportunities for complementing them with content, context, and meaning, but at the same time make it difficult to transfer them into practice. A more specific competence framework should not be shared across different lines of industries. The experience in applying different competence models shows that each can be successful in certain conditions [59, 60].

The development of a competency standard is popular among state [34, 55], nongovernmental [72], and non-profit organizations [67]. There are different frameworks for computer, information, or digital competence, as well as standards for digital literacy. Standards appeared in the definitive context, for example "ICT Competency Framework for Teachers" [72]. The analysis of a variety of frameworks [34, 55, 67, 72] demonstrates that they inherit the knowledge and skills requirements of the previous stage of IT development. It is notable that the standards often replace obsolete terms with modern ones. The "information era" was replaced by "digital era" in subtitles of [66, 67]. Since 2011, the "Skills Framework for the Information Age" [66] has been revised gaining new competencies; it has also expanded by 25% by 2021 [67]. It is obvious that in the context of rapid technological progress affecting a huge number of industries, the quantity and quality of competencies are too great to be fixed by a single solid framework.

The essence of digital competencies is addressed through personnel training in different business dimensions, for example education [7, 52], healthcare [38], finance [42], etc. In this case, the analysis of professional activities reflects its changes, but does not fully unleash DT's potential for the development and upgrading of professional performance, and its greater efficiency. Specifically, new technological opportunities are treated as applicable to immediate problems of the current information and technology paradigm. For example, assessing the independence of a student's performance in an educational task or the emerging problems, such as personalizing the content of an educational program are addressed in terms of technical capabilities of the industry. The main shortcoming of this approach is in an individual review of each business dimension, which does not adequately assess DC and their demand in society.

Traditionally, the higher education system is the main area of competence development and dissemination. The DC, in the context of university, is fully described [81]. The DC in higher education has great importance for lecturers and students' achievements in learning outcomes. Usually, the researchers [31, 51, 69] focusing on university context emphasize the needs and challenges of higher education, but not of society as a whole.

The leadership in the development and improvement of new DT belongs to businesses that can invest. The analysis of publications and the patenting activity [77] concludes that the academic community ensures the development of fundamental theoretical

and methodological propositions for DT, while their practical use enjoys intensive and successful growth in the business sector. The companies include the list of competencies they need, listing the expected hiring specifications in job advertisements. Given this, a common approach to explaining the quality of digital competencies is based on the analysis of job advertisements published by the companies. New technologies aimed at importing web-based content from digital platforms allow for analytical processing of almost all published job advertisements to identify general employers' requirements for the applicants, considering the field of expertise, job tenure, and other parameters [39]. The results achieved under this approach are indicative of the current demands of the labor market. Employer companies do not explicitly formulate a request for skilled personnel for the next ten-fifteen years, given the rapid digital environment changes [8]. Besides, hired labor does not cover all professional activities; more people find ways for professional development in entrepreneurship or freelancing [41]. Labor market requirements do not fully reflect the professional activity changes and do not define the DC of experts and professional employees.

A widespread method to identify the competencies relevant in the future is Foresight [29, 33], which allows for departing from technological concepts and nature of future activities, not from existing practice and needs. The *Atlas of Emerging Jobs*, compiled in 2014 [5], was based on the Foresight Method and predicted the emergence of new categories of jobs in the education sector not later than 2020. At the end of 2021, after most teachers had to give classes in a digital environment due to the COVID19 pandemic, it can be safely said that the mentioned professions have not yet appeared. However, tutoring and facilitating game-based or project-based learning have become common practice for the teaching profession. The comparison of the first edition of the *Atlas of Emerging Jobs* [5] with the third one [4] shows that it lost two professions and attained eight new professions in the education area by 2020. The analysis of the description and structure of the competencies for emerged professions indicates changes in existing professions. Moreover, the advancements in technology and the emergence of new opportunities change many professions' nature [35, 36].

In spite of different interpretations of competencies and views on it, contemporary researchers emphasize the high relevance of digital competencies for social and economic development [20–22, 70]. We can assume that new competencies have appeared in standards and frameworks post factum. The training system of higher education urgently demands the development of a flexible framework for the competencies to change industries and society. In this study, we consider digital competencies as the summation of industrial and technological knowledge, which cannot be limited by a solid framework to fix the quantity and quality of DC.

2 RESULTS AND DISCUSSION

Digital technologies, as next-generation IT, are becoming the major engine of growth in present-day society, and accessing them is a prerequisite for its economic development. Digital age differs from the informational stage in its extent of reliance on IT, which involves not only the IT owned by one enterprise, but almost all IT available in society, including data centers of the companies and individual smartphones. The use of various devices by people,

mobile Internet, the spread of the broadband Internet connection, Internet of Things - all this leads to the development of a new digital environment for business, science, public administration, and society at large. The digitalization of society provides the basis for sustainable development, which relies on the effective use of DT.

The study of DT [16] uses sound experience in the economy shows how new principles and standards of information and data processing implemented through DT have led to revolutionary changes in production, labor, and the system of economic relations. DT have replaced priority production technologies required for the manufacturing and delivery of any product or service. The proportion of tasks related to new information processing and non-trivial solutions to problems increased, while the number of conventional cognitive tasks and physical labor cope decreased [43]. The labor market analysis of the US economy shows that from 1960 to 2009, the country faced a reallocation of the economically active population from material production and agricultural sectors to service industries, healthcare, education, etc. Digitalization, infiltrating almost all industries, changes the nature of work even faster than in previous periods. The rate of employment sector changes is seen as a source of social risks, including mounting unemployment or declining living standards [3, 71]. Previous IT penetration experience has demonstrated that new technologies create more jobs than they have destroyed [80]. A similar conclusion has been reached by other researchers when they assessed the risks of unemployment in the digital age [6].

The EC (European Commission) has updated several practical competency frameworks [75]. These include DigComp (Digital Competence Framework for Citizens), e-CF (e-Competence Framework), etc. DigComp 2.0 has key components in 5 areas and 21 competencies presented in Table 1. The major usage of DigComp suggested by the EC includes employment services, job seekers, e-commerce, teachers, and learners. As the framework's title goes, the DigComp framework is sought for citizens' digital capacity, not for industry-level competencies.

Another approach for describing competence is The European e-Competence Framework (e-CF) [27]. The e-CF had been developed for more than ten years, and the e-CF 3.0 became a European standard and was officially published as the European Norm 16234 in 2016. The 41 competencies of the framework are classified according to five main IT business areas and relate to the European Qualifications Framework. The five areas are PLAN-BUILD-RUN-ENABLE-MANAGE in sequence. Though the taxonomy of competence elements in the five areas seems controversial; For example, "(A9) Innovating" can also be in the "BUILD" and "RUN" areas as well. The strength of the competence framework is used in practitioners and countries for comparative study. Furthermore, compared to DigComp, e-CF focuses more on the IT industry. The DigComp framework might be related to a personal level digital competence whereby the e-CF competence framework is only by IT industries prerequisites. For this reason, applying the two frameworks to the national level, digital competence studies need more elaborations. Furthermore, this need for conceptualizing and measuring national-level digital competence is indeed the starting point of a new integrated approach, one of the motivations for the current study.

Table 1: DigComp dimensions for digital competencies [75]

Competence Areas	Competencies
1. Information and data literacy	1.1 Browsing, searching and filtering data, information and digital content 1.2 Evaluating data, information and digital content 1.3 Managing data, information and digital content
2. Communication and collaboration	2.1 Interacting through digital technologies 2.2 Sharing through digital technologies 2.3 Engaging in citizenship through digital technologies 2.4 Collaborating through digital technologies 2.5 Netiquette 2.6 Managing digital identity
3. Digital content creation	3.1 Developing digital content 3.2 Integrating and re-elaborating digital content 3.3 Copyright and licenses 3.4 Programming
4. Safety	4.1 Protecting devices 4.2 Protecting personal data and privacy 4.3 Protecting health and well-being 4.4 Protecting the environment
5. Problem solving:	5.1 Solving technical problems 5.2 Identifying needs and technological responses 5.3 Creatively using digital technologies 5.4 Identifying digital competence gaps

In the digital age, the business course involves implementing DT and special software, but it is not a sufficient condition for sustainable development. For instance, technological expansion in the society of the 20th century was promising for productivity growth. However, the scientists identified a “productivity paradox” when IT developments led to a slowdown in productivity [17]. Moreover, some cases have negative effects when IT overlaps with existing business processes and slows them down [48]. The downside of society’s digital transformation is indicative of the “digital crisis” [24] when the business is forced to increase their DT investments, but there is no rise of efficiency.

Digital society’s significant economic benefits as a basis of sustainability are achieved by creating the innovations involving DT. Pure technologies are not enough for the digital transformation of the industries. It is necessary to find solutions to an urgent problem by applying technologies or creating new value. Competence as knowledge and skills should be considered as characteristic of the subject. The perception of competence as a personal attribute is common to different approaches to its definition. On the one hand, competence is considered a “basic quality of an individual,” formed from a unique combination of personal qualities, motives, ideas, and values with the accumulated knowledge and skills [68]. On the other hand, competence can characterize a group of people, a team, an organization, or an industry reflected in the organizational knowledge management theory [57]. The implementation of almost any activity in present-day conditions requires team efforts, a combination of knowledge and skills of several people, which results in the organization’s ability to provide training and education, be engaged in scientific activities, consulting, etc. In organizational

knowledge management theory, similar to individual competencies, the focal point is organization values [56] that influence an organization’s target setting and mission. The different types of subjects depend on scale and scope of accessing and possessing DT, way of use and benefit from DT. The types of subjects of DT and difference between them are shown in Table 2.

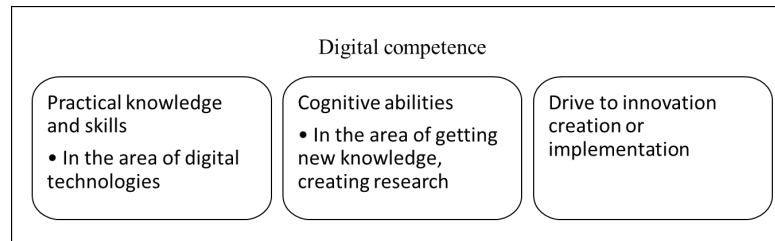
The representatives of business [10] and academic communities suggest that digitalization of the organization to mean a transformation whereby the role of the locomotive of change is played by complementary assets of computer capital [1, 23], for which unique industry expertise is a top priority. Economic and social factors, rather than technical factors, are key to digital transformation’s economic efficiency [28, 53]. There can be no meaningful economic effects in business operations only by transferring existing business processes to the digital environment. Innovations generated by people with the right industry and digital competencies are key to capturing social gains from technological progress.

So, sustainable development supposes the people and companies to possess the competencies to create innovations relied on DT and digitalization of society. The DC are changing from technical knowledge and skills of using DT forward to competencies of innovative activities with DT support in industries.

The studies on competence models suggest that the essence of competencies and their requirements are determined by the context [37, 76], namely, the subject matter area, industry, or line of business, to which they are applied. Acquiring knowledge of DT is very important and serves as a necessary condition for their successful application, but not sufficient to achieve a high level of efficiency in industry.

Table 2: Types of subjects of digital competencies in digital era

Character	Subject Type: Person	Subject Type: Industry	Subject Type: Society
Scale and Scope of DT use	Use own DT and devices to get access to digital services and platform	Use own or cloud facilities to involve the users' devices into its digital business model through APIs, Platforms, Clouds	Infrastructure for DT in society
Benefits from DT use	Extraction the personal advantages of digitalization in everyday and professional life (comfort, safety, consumption, employment)	Leadership in digital age, get new type of assets (big data, data flow) new type of instruments to improve performance, create innovations	Sustainability for society as whole (employment, wellbeing)
DCs Evolution	Apply DT for everyday duties Apply DT in professional duties (production, management, teaching, etc.) Entrepreneurship and innovation with DT	Apply DT in the existing processes and models Apply DT to create innovations Apply DT to achieve outstanding effects (sustainable growth)	Providing society with an infrastructure for implementation and use of personal and business DT Providing public services people and companies Providing institutions and support for sustainable development

**Figure 1: The concept of digital competence**

The development of digital competencies does not replace training in DT but complements it with skills, knowledge, and self-motivation to solve problems that ensure economic efficiency growth. The digitalization of society opens another dimension in competence models of professionals – the ability to create innovative solutions by making use of DT. The area of expertise should be included in the definition of digital competence and its mandatory components (cognitive, practical, motivational). Figure 1 demonstrates the key elements of DT. By digital competence we mean the ability of an individual or organization to solve non-routine tasks using digital technology or a combination of them in certain business activity or industry.

The action-oriented component in digital competence allows the practitioner to formulate problems for DT application or directly apply DT to business operations. At the same time, subject matter area expertise and relevant skills will play a key role in digital transformation of industry. For example, artificial intelligence as a set of technologies creates new opportunities for improving economic and management processes; however, the content of these processes varies significantly between industries, impeding the transfer of competencies between them. The competencies required for integrating artificial intelligence into healthcare will differ from those needed in the metallurgical industry since both the processes and technologies will be determined by industry-specific issues, legal regulations, or the level of accountability for the results.

Cognitive ability in the digital competence structure is in charge of obtaining new knowledge necessary to create innovation. Routine operations in a digital age are performed automatically, and the tasks that require non-trivial solutions, which are beyond professional patterns and stereotypes, remain within the competence of an expert. The creative activity of people in a digital world involves the competencies identified by the OECD as transformative competencies [58], but it also requires digital and subject matter expertise. By applying creative and entrepreneurial initiatives, a person or organization uses DT to generate innovation, ensuring improved business performance.

The Internet rapidly distributes digital innovation, providing global reach. There is no need for training to use the power of DT in everyday life to order goods and services, access to online entertainment, or supporting IoT for housekeeping. But the DC to transform the industries are urgent to the tasks of sustainable development of society. The potential of the digital age for sustainable development is unevenly revealed in industries. The need to possess not only access to technologies, but also unique knowledge of the subject area limits the direct transfer of technologies and competencies between industries without their adaptation or modification. The variety of ways to use digital technologies to solve economic tasks poses the problem of forming the digital competencies and designing a system for the dissemination and sharing of competencies. Implications of the research and future research

direction in this domain are highlighted for academia and business practitioners.

3 CONCLUSION

The conclusion of this study of digital competence conceptualization can be summarized as follows:

Changes in the information and technology paradigm of socioeconomic development lead to significant transformations of job assignments, manifested in the reduction of routine and algorithm-driven operations and an increase in challenging cognitive tasks. The digitalization of society should intensify the innovative potential of business entities and serve as a guide to sustainable development. Additionally, the benefits of DT application, as of any other technology, depend on the practitioners' competencies.

Digital competencies are not extension of computer and information literacy since the use of digital technologies is entirely different from the experience of user with a personal computer or web services. Digital competencies should ensure new economic benefits for practitioners or productivity growth based on digital resources. Access to digital technologies and digital competencies to address industry challenges determine the success of digital transformation. The mechanisms of digital transformation include innovations, such as material changes of business processes, models, creation of digital products and markets, etc. Creativity is one of the components of digital competence, next to practical knowledge and a drive to innovate.

The definition of digital competence consists of two main components: subject matter expertise and awareness of digital technologies' potential. It is hardly possible to define a set of multifunctional digital competencies that will be equally demanded in all industries. The practitioner's skills and unique subject matter or industry expertise are crucial for applying digital competencies for economic advancement. In fact, digital competence is the capability to innovate using the digital opportunities of the industry.

For each field of business, there is a set of competencies required to work with a certain DT or a set of them, which allows for the development of a variety of digital competencies. The number of possible digital competencies can be estimated based on the number of industry challenges that can be resolved by applying DT. Given the rapid improvement and creation of new DT, the number of competencies is not fixed. If you map a set of competencies into a matrix, they can be studied utilizing mathematical analysis. The further quantitative research of DC will support the development of ways to manage DC as a key driver of digital transformation of industries.

There is a huge potential for further research to identify the quantity and quality of DC for different types of subjects (person, industry, society). Sustainable development needs the solution of this scientific question as one of its drivers.

REFERENCES

- [1] Ananyin, V.I., Zimin, K.V., Lugachev, M.I., Gimranov, R.D., Skripin, K.G. 2018. Digital organization: Transformation into the new reality. *Business Informatics*, no. 2 (44), pp. 45–54. DOI: 10.17323/1998-0663.2018.2.45.54.
- [2] Arias-Pérez, J., Velez-Ocampo, J. and Cepeda-Cardona, J. 2021. Strategic orientation toward digitalization to improve innovation capability: why knowledge acquisition and exploitation through external embeddedness matter. *Journal of Knowledge Management*, Vol. 25 No. 5, pp. 1319–1335. DOI: 10.1108/JKM-03-2020-0231
- [3] Arntz, M., T. Gregory and U. Zierahn 2016. The Risk of Automation for Jobs in OECD Countries: A Comparative Analysis. OECD Social, Employment and Migration Working Papers, No. 189, OECD Publishing, Paris, <https://doi.org/10.1787/5jlz9h56dvq7-en>.
- [4] Atlas of Emerging Jobs 3.0 2020. Atlas novyh professiy 3.0. edited by D. Varlamova and D. Sudakova. Moscow, Intelligence Literature. Retrieved January 5, 2022 from https://atlas100.ru/upload/pdf_files/atlas.pdf (In Russ.)
- [5] Atlas of Emerging Jobs. 2014. Atlas novyh professiy. Moscow. SKOLKOVO Moscow School of Management. Retrieved January 5, 2022 from <https://files.asi.ru/iblock/ce8/ce82467fa9a2b9d99d1314475b297c3e/Atlas.pdf> (In Russ.)
- [6] Autor, D. and Salomons, A. 2017. Does productivity growth threaten employment? //ECB Forum on Central Banking, Sintra, Portugal. Retrieved January 5, 2022 from https://www.ecb.europa.eu/pub/conferences/shared/pdf/20170626_ecb_forum/D_Autor_A_Salomons_Does_productivity_growth_threaten_employment.pdf
- [7] Badiozaman, I. F. A., Leong, H. J. and Wong, W. 2020. Embracing educational disruption: a case study in making the shift to a remote learning environment. *Journal of Applied Research in Higher Education*. DOI: 10.1108/JARHE-08-2020-0256
- [8] Bessen, J.E. 2016. How computer automation affects occupations: Technology, jobs, and skills. Boston University School of Law, Law and Economics Research Paper No. 15–49. Boston, MA: Boston University.
- [9] Blewitt, J. 2012. Understanding sustainable development. Routledge.
- [10] Boulton, C. 2021. 10 most misused buzzwords in IT. CIO. United States. Retrieved January 5, 2022 from <https://www.cio.com/article/3604596/most-misused-buzzwords-in-information-technology.html>
- [11] Box, S. and Lopez-Gonzalez, J. 2017. The future of technology: Opportunities for ASEAN in the digital economy. *Global megatrends: Implications for the ASEAN economic community*, pp.37–60.
- [12] Bratianu, C., Hadad, S. and Bejinaru, R. 2020. Paradigm shift in business education: A competence-based approach. *Sustainability*, 12(4) doi:10.3390/su12041348
- [13] Bresnahan, T. F. and Trajtenberg, M. 1995. General purpose technologies' Engines of growth?. *Journal of econometrics*, 65(1), pp. 83–108.
- [14] Broadband Commission for Sustainable Development 2017. Digital skills for life and work. Retrieved January 5, 2022 from https://www.skillsforemployment.org/KSP/en/Details/?dn=WCMSTEST4_193296
- [15] Brundtland, G. H. 1987. Selected speeches on the Commission and its report by HE Mrs. Gro Harlem Brundtland, Prime Minister of Norway and Chairman of the World Commission on Environment and Development. WCED archive collection; v. 42, doc. 1–36.
- [16] Brynjolfsson, E. and Saunders, A. 2010. *Wired for innovation: How information technology is reshaping the economy*. Cambridge, London, MIT Press.
- [17] Brynjolfsson, E., Rock, D. and Syverson, C. 2021. The productivity J-curve: How intangibles complement general purpose technologies. *American Economic Journal: Macroeconomics*, 13(1), pp. 333–72. DOI: 10.1257/mac.20180386
- [18] Carlucci, D., Mura, M. and Schiuma, G. 2020. Fostering employees' innovative work behaviour in healthcare organisations. *International Journal of Innovation Management*, 24(2) doi:10.1142/S1363919620500140
- [19] Carretero, S., Vuorikari, R. and Punie, Y. 2017. The digital competence framework for citizens Publications Office of the European Union. <https://ec.europa.eu/jrc/en/publication/eur-scientific-and-technical-research-reports/digcomp-21-digital-competence-framework-citizens-eight-proficiency-levels-and-examples-use>, Accessed Jan 2021
- [20] Carretero, S.; Vuorikari, R. and Punie, Y. 2017. DigComp 2.1: The Digital Competence Framework for Citizens with eight proficiency levels and examples of use, EUR 28558 EN, doi:10.2760/38842
- [21] Castelfranchi, C. 2007. Six critical remarks on science and the construction of the knowledge society. *Journal of Science Communication*, 6 (4)
- [22] Cebrián, G. and Junyent, M. 2015. Competencies in education for sustainable development: Exploring the student teachers' views, *Sustainability*, 7 (3), pp. 2768–2786
- [23] Coskun-Setirek, A. and Tanrikulu, Z. 2021. Digital innovations-driven business model regeneration: A process model. *Technology in Society*, 64, 101461. <https://doi.org/10.1016/j.techsoc.2020.101461>
- [24] Dneprovskaya, N. 2019. The digitalization crisis in the knowledge economy. Proceedings of the International Conference on Intellectual Capital, Knowledge Management and Organisational Learning, ICICKM, 2019–December, pp. 115–122. DOI: 10.34190/IKM.19.107
- [25] Dneprovskaya, N.V. and Shevtsova, I.V. 2021. Methodological Approach to Use of Web Content by Small Business. *Scientific and Technical Information Processing*, 48(2), pp.78–86. Doi:10.3103/S0147688221020040
- [26] Endres, H., Huesig, S. and Pesch, R. 2021. Digital innovation management for entrepreneurial ecosystems: services and functionalities as drivers of innovation management software adoption. *Review of Managerial Science*, 1–22. DOI: 10.1007/s11846-021-00441-4
- [27] European Committee for Standardization. 2021. CEN/TC 428 - ICT Professionalism and Digital Competences. Retrieved January 5, 2022 from https://standards.cen.eu/dyn/www/f?p=204:32:0:::FSP_ORG_ID,FSP_LANG_ID:1218399,25&cs=10D4962B00BCD3044A1F1BF7BA26EDD21

- [28] Evans, N. and Price, J. 2020. Development of a holistic model for the management of an enterprise's information assets. *International Journal of Information Management*, 54, 102193. DOI: 10.1016/j.ijinfomgt.2020.102193
- [29] Fossen, F. and Sorgner, A. 2019. Mapping the Future of Occupations: Transformative and Destructive Effects of New Digital Technologies on Jobs. *Foresight and STI Governance*, vol. 13, no 2, pp. 10–18. DOI: 10.17323/2500-2597.2019.2.10.18.
- [30] Gavrilova, T.A., Alsufyev, A.I. and Grinberg, E.Y. 2017. Knowledge visualization: Critique of the St. Gallen School and an analysis of contemporary trends. *Business Informatics*, no. 3 (41), pp. 7–19. DOI: 10.17323/1998-0663.2017.3.7.19
- [31] Hatlevik, O. E., Throndsen, I., Loi, M. and Gudmundsdottir, G. B. 2018. Students' ICT self-efficacy and computer and information literacy: Determinants and relationships. *Computers & Education*, 118, 107–119. DOI: 10.1016/j.compedu.2017.11.011
- [32] Hilbert, M. and López, P. 2011. The world's technological capacity to store, communicate, and compute information. *science*, 332(6025), 60–65. DOI: 10.1126/science.1200970
- [33] Hines, A. 2019. Getting Ready for a Post Work Future. *Foresight and STI Governance*, vol. 13, no 1, pp. 19–30. DOI: 10.17323/2500-2597.2019.1.19.30.C.20
- [34] House of Lords. 2017. Digital Skills in the United Kingdom. House of Lords Library Briefing. Retrieved January 5, 2022 from https://www.legco.gov.hk/general/english/library/stay_informed_overseas_policy_updates/digital_skills_in_the_uk.pdf
- [35] Houston Foresight Program. 2014. The future of student needs: 2025 and beyond, Indianapolis, IN: Lumina Foundation. Retrieved January 5, 2022 from <https://www.houstonforesight.org/wp-content/uploads/2015/05/FutureStudentNeeds2025LR.pdf>
- [36] Houston Foresight Program. 2017. The future of work 2050 for NASA LARC, Houston, TX: University of Houston. Retrieved January 5, 2022 from <https://www.houstonforesight.org/wp-content/uploads/2017/10/NASA-LaRC-Future-of-Work-White-Paper.pdf>
- [37] Janssen, J., Stoyanov, S., Ferrari, A., Punie, Y., Pannekeet, K. and Sloep, P. 2013. Experts' views on digitalcompetence: Commonalities and differences. *Computers & Education*, 68, 473–481. <https://doi.org/10.1016/j.compedu.2013.06.008>
- [38] Jimenez, G., Spinazze, P., Matchar, D., Huat, G. K. C., van der Kleij, R. M., Chavannes, N. H. and Car, J. 2020. Digital health competencies for primary healthcare professionals: A scoping review. *International Journal of Medical Informatics*, 104260. DOI: 10.1016/j.ijmedinf.2020.104260
- [39] Kim, J., Warg, E. and Moen, W. 2013. Competencies required for digital curation: An analysis of job advertisements. DOI: <https://doi.org/10.2218/ijdc.v8i1.242>
- [40] Kudryavtsev, D. and Sadykova, D. 2019. Towards Architecting a Knowledge Management System: Requirements for an ISO Compliant Framework. In *IFIP Working Conference on The Practice of Enterprise Modeling* (pp. 36–50). Springer, Cham.
- [41] Kuzminov, Ya., Sorokin, P., and Froumin, I. 2019. Generic and Specific Skills as Components of Human Capital: New Challenges for Education Theory and Practice. *Foresight and STI Governance*, vol. 13, no 2, pp. 19–41. DOI: 10.17323/2500-2597.2019.2.19.41
- [42] Lavrinenko, A. and Shmatko, N. 2019. Twenty-First Century Skills in Finance: Prospects for a Profound Job Transformation. *Foresight and STI Governance*, vol. 13, no 2, pp. 42–51. DOI: 10.17323/2500-2597.2019.2.42.51
- [43] Levy, F. and Murnane, R.J. 2013. *Dancing with robots: Human skills for computerized work*. Washington, DC: Third Way NEXT. Retrieved January 5, 2022 from <https://www.thirdway.org/report/dancing-with-robots-human-skills-for-computerized-work>
- [44] Linkov, Igor, Benjamin D. Trump, Kelsey Poinette-Jones, and Marie-Valentine Florin. 2018. Governance Strategies for a Sustainable Digital World. *Sustainability* 10, no. 2: 440. <https://doi.org/10.3390/su10020440>
- [45] Lissitsa, S., Chachashvili-Bolotin, S. and Bokek-Cohen, Y. A. 2017. Digital skills and extrinsic rewards in late career. *Technology in Society*, 51, 46–55. <https://doi.org/10.1016/j.techsoc.2017.07.006>
- [46] Lloyd, A. and Talja, S. (Eds.). 2010. *Practising information literacy: Bringing theories of learning, practice and information literacy together*. Elsevier.
- [47] Lukas, M., Bem-Haja, P., Siddiq, F., Moreira, A. and Redecker, C. 2021. The relation between in-service teachers' digital competence and personal and contextual factors: What matters most? *Computers and Education*, 160, 104052. <https://doi.org/10.1016/j.compedu.2020.104052>
- [48] Mankins, M. 2016. Is technology really helping us get more done. *Harvard Business Review*, 25. Retrieved January 5, 2022 from <https://hbr.org/2016/02/is-technology-really-helping-us-get-more-done>
- [49] Martin, A. and Grudziecki, J. 2006. DigEuLit: Concepts and Tools for Digital Literacy Development, Innovation in Teaching and Learning in Information and Computer Sciences, 5:4, pp. 249–267. DOI: 10.11120/ital.2006.05040249
- [50] McAfee, A., and Brynjolfsson, E. 2017. *Machine, platform, crowd: Harnessing our digital future*. WW Norton & Company. P. 402
- [51] Mehrvarz, M., Heidari, E., Farrokhi, M. and Noroozi, O. 2021. The mediating role of digital informal learning in the relationship between students' digital competence and their academic performance. *Computers & Education*, 167, 104184. DOI: 10.1016/j.compedu.2021.104184
- DOI: 10.1016/j.compedu.2021.104184
- [52] Mezentseva, D.A., Dzhavakh, E.S., Eliseeva, O.V., Bagautdinova, and A.Sh. 2020. On the Question of Pedagogical Digital Competence. *Vyshee obrazovanie v Rossii = Higher Education in Russia*. Vol. 29, no. 11, pp. 88–97. (In Eng., abstract in Russ.) DOI: <https://doi.org/10.31992/0869-3617-2020-29-11-88-97>
- [53] Moazed, A. and Jonson, N. L. 2016. *Modern Monopolies. What It Takes to Dominate the 21st Century Economy*. St. Martin's Press. New York.
- [54] Narula, S., Prakash, S., Dwivedy, M., Talwar, V., and Tiwari, S. P. 2020. Industry 4.0 adoption key factors: an empirical study on manufacturing industry. *Journal of Advances in Management Research*. DOI: 10.1108/JAMR-03-2020-0039
- [55] National Competence Standards. 2021. Classification of Industry Competence. Retrieved January 5, 2022 from <https://www.ncs.go.kr/th01/TH-102-001-02.scd0> (in Korean)
- [56] Nonaka, I. and Toyama, R. 2015. The knowledge-creating theory revisited: knowledge creation as a synthesizing process. In *The essentials of knowledge management* (pp. 95–110). Palgrave Macmillan, London.
- [57] Nonaka, I., Toyama, R. and Hirata, T. 2008. *Managing flow: a process theory of the knowledge-based firm* Published by Palgrave Macmillan, Houndmills, 2008, 252 pp. ISBN: 978-0-230-55376-7
- [58] OECD. 2018. The Future of Education and Skills: Education 2030. Position paper, Retrieved January 5, 2022 from [https://www.oecd.org/education/2030/E2030%20Position%20Paper%20\(05.04.2018\).pdf](https://www.oecd.org/education/2030/E2030%20Position%20Paper%20(05.04.2018).pdf)
- [59] Pellicer, L. O., Anderson, L. W. 2001. Teacher Leadership: A Promising Paradigm for Improving Instruction in Science and Mathematics. Retrieved January 5, 2022 from <https://eric.ed.gov/?id=ED465586>
- [60] Pettersson, F. 2018. On the issues of digital competence in educational contexts—a review of literature. *Education and information technologies*, 23(3), pp. 1005–1021. DOI: 10.1007/s10639-017-9649-3
- [61] Rychen, D.S. 2016. Working paper. Education 2030: Key competencies for the future. OECD, Paris. Retrieved January 5, 2022 from <http://www.oecd.org/education/2030/E2030-CONCEPTUAL-FRAMEWORK-KEY-COMPETENCIES-FOR-2030.pdf>
- [62] Sachs, J., Kroll, C., Lafortune, G., Fuller, G. and Woelm, F. 2021. *Sustainable development report 2021*. Cambridge University Press.
- [63] Schiuma, G. and Carlucci, D. 2018. Managing strategic partnerships with universities in innovation ecosystems: A research agenda. *Journal of Open Innovation: Technology, Market, and Complexity*, 4(3) doi:10.3390/joitmc4030025
- [64] Schmitz, A. P. and Foelsing, J. 2018. Social Collaborative Learning Environments: A Means to Reconceptualise Leadership Education for Tomorrow's Leaders and Universities?. In *The Disruptive Power of Online Education*. Emerald Publishing Limited.
- [65] Shevtsova, I. and Dneprovskaya, N. 2021. Transformation of the Digital Environment into Actor of E-Learning. 2021 International Conference on Quality Management, Transport and Information Security, Information Technologies (IT&QM&IS), pp. 846–849, doi: 10.1109/ITQMIS53292.2021.9642745.
- [66] SFIA Foundation. 2011. SFIA 5 framework reference: Skills defined in categories and subcategories. Retrieved January 5, 2022 from <https://sfia-online.org/en/legacy-sfia/sfia-5/sfia5ref-en-r4.pdf/view>
- [67] SFIA Foundation. 2021. Guidance for the use of the SFIA standard: The global skills and competency framework for the digital world. Retrieved January 5, 2022 from <https://sfia-online.org/en/sfia-8/documentation/about-sfia-v8-0-about-sfia-en-210928.pdf/view>
- [68] Spencer, L. M. and Spencer, P. S. M. 2008. *Competence at Work models for superior performance*. John Wiley & Sons.
- [69] Teo, T., Unwin, S., Scherer, R. and Gardiner, V. 2021. Initial teacher training for twenty-first century skills in the Fourth Industrial Revolution (IR 4.0): A scoping review. *Computers & Education*, 104223. DOI: 10.1016/j.compedu.2021.104223
- [70] Tirado-Morueta, R., Aguaded-Gómez, J. I. and Hernando-Gómez, Á. 2018. The socio-demographic divide in Internet usage moderated by digital literacy support. *Technology in Society*, 55, 47–55. <https://doi.org/10.1016/j.techsoc.2018.06.001>
- [71] Tudoran, C. 2020. New Training Trends in High-Risk Technology Fields. In *Proceedings of the International Conference on Business Excellence* (Vol. 14, No. 1, pp. 474–489). Scienco. DOI: 10.2478/picbe-2020-0045
- [72] UNESCO. 2018. *ICT Competency Framework for Teachers*. Retrieved January 5, 2022 from <https://unesdoc.unesco.org/ark:/48223/pf0000265721>
- [73] UNESCO. 2022. Sustainable Development. Retrieved January 5, 2022 from <https://en.unesco.org/themes/education-sustainable-development/what-is-esd/sd>
- [74] van Tonder, C., Schachtebeck, C., Nieuwenhuizen, C. and Bossink, B. 2020. A framework for digital transformation and business model innovation. *Management: Journal of Contemporary Management Issues*, 25(2), pp. 111–132. DOI: 10.30924/mjcmi.25.2.6
- [75] Vuorikari, R., Punie, Y., Carretero Gomez S. and Van den Brande, G. 2016. *DigComp 2.0: The Digital Competence Framework for Citizens. Update Phase 1: The Conceptual Reference Model*. Luxembourg Publication Office of the European Union. EUR 27948 EN. doi:10.2791/11517
- [76] Winterton, J., Delamare-Le Deist, F. and Stringfellow, E. 2006. *Typology of knowledge, skills and competencies: clarification of the concept and prototype*. Luxembourg: Office for Official Publications of the European Communities.

- [77] WIPO. 2021. Patentscope database. Retrieved January 5, 2022 from <https://patentscope.wipo.int/>
- [78] World Bank. 2016. Digital Adoption Index. Retrieved January 5, 2022 from <https://www.worldbank.org/en/publication/wdr2016/Digital-Adoption-Index>
- [79] World Economy Forum. 2017. The Global Human Capital Report. Retrieved January 5, 2022 from <https://www.weforum.org/reports/the-global-human-capital-report-2017>
- [80] Zemtsov, S. 2017. Robots and potential technological unemployment in the Russian regions: Review and preliminary results. *Voprosy Ekonomiki*, 7, pp. 142-157. (In Russ.) <https://doi.org/10.32609/0042-8736-2017-7-142-157>
- [81] Zhao, Y., Pinto Llorente, A.M., and Sánchez Gómez, M.C. 2021. Digital competence in higher education research: A systematic literature review, *Computers & Education*, Vol. 168, 104212, <https://doi.org/10.1016/j.compedu.2021.104212>.